

Review retrospective:
the affine- κ_3 asymptotic — closing the anharmonic arc

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Abstract

A soundness review of `Laplace/OneD/AnharmonicKappa3Affine.lean`: the affine generalisation of the anharmonic third-cumulant asymptotic — for affine observables $(a_i x + b_i)$, $t^2 \kappa_3 \rightarrow a_1 a_2 a_3 \cdot (-\alpha/\lambda^3)$. Result: **a clean fidelity pass with six dead-code removals** — two dead `opens` (`Filter`, `Topology`) and four dead proof-locals. This is the last anharmonic file; the arc is now reviewed end-to-end.

1 Setting

The affine lift of `kappa3_anharmonic_id_id_id_asymptotic`: it shows the joint third cumulant on affine observables reduces to the scalar `id, id, id` case times $a_1 a_2 a_3$ (the constants b_i vanish on constant cumulant slots, the multipliers pull through by trilinearity), then composes with the base $t^2 \kappa_3[\text{id}] \rightarrow -\alpha/\lambda^3$. It is the final unreviewed file of the anharmonic stack.

2 What was reviewed

The headline `kappa3_anharmonic_shifted_affine_asymptotic` and its supporting private infrastructure (the affine-to-scalar identity, the cubic-moment expansion `gibbsExp_anharmonic_cubic_eq`, and the `polynomial × exp` integrability lemmas).

3 Fidelity / soundness findings

Sound; no defects. The affine-to-scalar reduction is proved concretely (not by an abstract trilinearity lemma): `kappa3_volume_unfold` expands κ_3 into the five-term cumulant formula, `gibbsExp_anharmonic_cubic_eq` expands each affine/product integrand into moments of $1, x, x^2, x^3$, and the closing `ring` cancels every b_i term, leaving $a_1 a_2 a_3$. The limit value $a_1 a_2 a_3 (-\alpha/\lambda^3)$ follows by scaling the base asymptotic and a `congr`’ along eventual- $t > 0$ equality. The reviewer confirmed all of this; the value matches the empirical $-\alpha/\lambda^3$ (e.g. -0.5 at $(\lambda, \alpha) = (1, 0.5)$). One report-only note: the private `kappa3_volume_unfold` docstring says the expansion is “independent of the perturbation direction A ”, which is strong as written (A survives in the observables); the operative meaning (the $h = 0$ weight/partition is A -independent) is defensible, so it was left.

4 Simplifications applied

Six dead-code removals. `open Filter` is dead — every filter name (`Filter.Tendsto`, `Filter.atTop`, `Filter.eventually_gt_atTop`, `Filter.Eventually.of_forall`) is written qualified, and `filter_upwards` is a tactic. `open Topology` is dead too: the file uses root-level `nhds`, never the bare $\mathcal{N}$ notation. `open MeasureTheory` stays (25 bare `∫/Integrable/volume`). And four dead locals: `h12` (an integrability `have` the later step-by-step construction never uses) and the three cubic-form rewrites `h_id_id_id`, `h_id_id`, `h_id` (which the `conv_rhs` re-proves inline instead of using). Build green at 8258 jobs, warning count 0, signatures byte-identical. GPT affirmed.

5 Disagreements and open questions

None.

6 Lessons

- **An asymptotics file can be Filter/Topology-open-free.** Even a `Tendsto ...atTop (nhds ...)` statement needs neither `open` when every filter name is written qualified and the limit point is `nhds` (not the $\mathcal{N}$ notation). The bare-name greps (no bare `atTop/Tendsto`, no bare $\mathcal{N}$) called both opens dead before the build confirmed it.
- **Inline-reproved rewrites strand their named `haves`.** The three `h_id...` cubic-form equalities were prepared as named `haves`, then the `conv_rhs` re-derived the same rewrites with inline `show ... from by ring` — leaving the named versions dead. When a proof both names a fact and re-proves it inline downstream, the named one is the dead one.

7 Follow-ups

- The anharmonic arc is now fully reviewed. Remaining `laplace OneD/`: the harmonic regularity/observable/perturbed/affine- κ_3 files and `IntegralRemainder` (2668L); plus `QuarticBoundedPrior/QuarticBoundedPriorTestFn`.
- Forward-tide / report-only: tighten the `kappa3_volume_unfold` “independent of A ” private docstring to “the $h = 0$ weight is A -independent”.